2. Construct CLR parser for the given grammar

$S \to L = R \mid R$

$L \to *R \mid id$

$R \to L$

Also the parse the string $id = id$

Solution

Augmented Grammar

$S' \to S$

$S \to L = R$

$S \to R$

$L \to *R$

$L \to id$

$R \to L$

Closure (I)

I_0
$$\begin{cases} S' \to \cdot S, \$ \\ S \to \cdot L = R, \$ \\ S \to \cdot R, \$ \\ L \to \cdot *R, =/\$ \\ L \to \cdot id, =/\$ \\ R \to \cdot L, \$ \end{cases}$$

$S' \to \cdot S, \$$
$\alpha B \beta, a$
first $[\varepsilon, \$] = \$$

$S \to \cdot \underset{B}{L} \underset{\beta}{= R}, \$$
FIRST $(= R, \$)$

$R \to \cdot L, \$$

Already we have to written the production for L, so we have to write only look ahead

$S' \to \cdot S, \$$
$S \to \cdot L = R, \$$
$S \to \cdot R, \$$
$\left\{ \begin{array}{l} L \to \cdot *R, = \\ L \to \cdot id, = \end{array} \right\}$ ①
$R \to \cdot L, \$$
$\left\{ \begin{array}{l} L \to \cdot *R, \$ \\ L \to \cdot id, \$ \end{array} \right\}$ ②

① & ② can be combined

$L \to \cdot *R, =, \$$
$L \to \cdot id, = \$$

Canonical LR(1) Item

goto (I_0, S)

I_1: $S' \to S \cdot, \$$

goto (I_0, L)

I_2: $S \to L \cdot = R, \$$
$R \to L \cdot, \$$ (R)

goto(I_0, R)

I_3: $S \to R\cdot, \$$ (R)

goto(I_0, *)

I_4: $L \to *\cdot R, =|\$$

$R \to \cdot L, =|\$$

$L \to \cdot *R, =|\$$

$L \to \cdot id, =|\$$

goto(I_0, id)

I_5: $L \to id\cdot, =|\$$ (R)

goto(I_2, =)

I_6: $S \to L=\cdot R, \$$

$R \to \cdot L, \$$

$L \to \cdot *R, \$$

$L \to \cdot id, \$$

goto(I_4, R)

I_7: $L \to *R\cdot, =|\$$ (R)

goto(I_4, L)

I_8: $R \to L\cdot, =|\$$ (R)

goto(I_4, *)

$L \to *\cdot R, =|\$$

$R \to \cdot L, =|\$$

$L \to \cdot *R, =|\$$

$L \to \cdot id, =|\$$

— I_4

goto(I_4, id)

$L \to id\cdot, =|\$$ — I_5

goto(I_6, R)

I_9: $S \to L=R\cdot, \$$ (R)

goto(I_6, L)

I_{10}: $R \to L\cdot, \$$ (R)

goto(I_6, *)

I_{11}: $L \to *\cdot R, \$$

$R \to \cdot L, \$$

$L \to \cdot *R, \$$

$L \to \cdot id, \$$

goto(I_6, id)

I_{12}: $L \to id\cdot, \$$ (R)

goto(I_{11}, R)

I_{13}: $L \to *R\cdot, \$$ (R)

goto(I_{11}, L)

$R \to L\cdot, \$$ — I_{10}

goto(I_{11}, *)

$L \to *\cdot R, \$$

$R \to \cdot L, \$$

$L \to \cdot *R, \$$

$L \to \cdot id, \$$

$\to I_{11}$

goto(I_{11}, id)

$L \to id\cdot, \$$ — I_{12}

Construction of CLR parsing table

States	ACTION				GOTO		
	id	=	*	$	S	L	R
I_0	s_5		s_4		1	2	3
I_1				Accept			
I_2		s_6		r_5			
I_3				r_2			
I_4	s_5		s_4			8	7
I_5		r_4		r_4			
I_6	s_{12}		s_{11}			10	9
I_7		r_3		r_3			
I_8		r_5		r_5			
I_9				r_1			
I_{10}				r_5			
I_{11}	s_{12}		s_{11}			10	13
I_{12}				r_4			
I_{13}				r_3			

Reduce

1. $S \to L = R$
2. $S \to R$
3. $L \to *R$
4. $L \to id$
5. $R \to L$

$I_3 : S \to R., \$$

$(3, \$) \to r_2$

State Action production rule

$I_5 : L \to id., =/\$$

$(5, (=/\$)) = r_4$

$I_7 : L \to *R., =/\$$

$(7, =/\$) = r_3$

CLR Parsing

id = id $

Stack	Input String	Action	
$0	id = id $	Shift S_5	(0, id)
$0id5	= id $	Reduce r_4 (L→id)	(5, =)
$0L2	= id $	Shift S_6	(2, =)
$0L2=6	id $	Shift S_{12}	(6, id)
$0L2=6id12	$	Reduce r_4 (L→id)	(12, $)
$0L2=6L10	$	Reduce r_5 (R→L)	
$0L2=6R9	$	Reduce r_1 (S→L=R)	(9, $)
$0S1	$	Accept	(1, $)

State Action

State Goto
0L → 2

$0L2=6id12
L
$0L2=6L
$0L2=6L10

$0L2=6R9
S→L=R
lengths
(3×2=6)
0\boxed{L2=6R9}$
$0S

3. Construct CLR parser for the grammar

$E \to E + T$
$E \to T$
$T \to id(E)$
$T \to id$

Solution:

Augmented grammar

$E' \to E$
$E \to E + T$
$T \to id(E)$
$T \to id$
$E \to T$

Closure (I)

~~$E' \to \cdot E, \$$~~ $E' \to \cdot E, \$$
~~$E' \to \cdot E + T,$~~ $E \to \cdot E + T, \$/+$
$E \to \cdot T, \$/+$
$T \to \cdot id(E), \$/+$
$T \to \cdot id, \$/+$

$E' \to \cdot \underline{E}, \$$ FIRST $[\epsilon, \$] = \$$
1 $E \to \cdot \underline{E} + T, \$$ FIRST $[\epsilon + T \$] = +$
$E \to \cdot \underline{T}, \$$ FIRST $T [\epsilon, \$] = \$$
$E \to \cdot E + T, +$ FIRST $T [\epsilon, +] = +$
2 $E \to \cdot \underline{T}, +$
$T \to \cdot id(E), \$, +$
$T \to \cdot id, \$, +$

$I_0 \to$

Combine 1 and 2

$E' \to \cdot E, \$$
$E \to \cdot E + T, \$/+$
$E \to \cdot T, \$/+$
$T \to \cdot id(E), \$/+$
$T \to \cdot id, \$/+$

Canonical LR(1) Item

I_0: E' → .E, $
E → .E+T, $/+
E → .T, $/+
T → .id(E), $/+
T → .id, $/+

goto(I_0, E)

I_1: E' → E., $
E → E.+T, $/+

[Check E in the RHS and move dot (.) to next position. No need to find FIRST() in this scenario]

goto(I_0, T)

I_2: E → T., $/+

goto(I_0, id)

I_3: T → ~~.id(E)~~ id.(E), $/+
T → id., $/+

goto(I_1, +)

I_4: E → E+.T, $/+
T → .id(E), $/+
T → .id, $/+

[After . check for non-terminal here we have T, add only the production of T not lookahead. After that find the lookahead for T → .id(E), T → .id from the parent production
E → E+.T, $,+
FIRST(T) = [, $/+]]

goto(I_3, ()

~~T → id.(E), $/+~~

I_5: T → id(.E), $/+
E → .E+T,)/+
E → .T,)/+
T → .id(E),)/+
T → .id,)/+

FIRST(E) = [), $/+] =)

① E → .E+T,)
E → .T,)

② E → .E+T, +
E → .T, +

FIRST(E) = [+T,)]

E → .E+T,)/+
E → .T,)/+
① + ②

E → .T,)/+ FIRST(T) = [,)/+]

goto(I_4, T)

I_6: E → E+T., $/+

goto(I_4, id)

I_3: $T \to id.(E)$, $/+
$T \to id.$, $/+

goto(I_5, E)

I_7: $T \to id(E.)$, $/+
$E \to E.+T$,)/+

goto(I_5, T)

I_8: $E \to T.$,)/+

goto(I_5, id)

I_9: $T \to id.(E)$,)/+
$T \to id.$,)/+

goto(I_7,))

I_{10}: $T \to id(E).$, $/+

goto(I_7, +)

I_{11}: $E \to E+.T$,)/+
$T \to .id(E)$, ~~$~~)/+
$T \to .id$,)/+

(

(

goto(I_{11}, T)

I_{15}: $E \to E+T.$,)/+

goto(I_9, ( )

I_{12}: $T \to id(.E)$,)/+
$E \to .E+T$,)/+
$E \to .T$,)/+
$T \to$ ~~id~~ $.id(E)$,)/+
$T \to .id$,)/+

goto(I_{11}, T)

I_8: $E \to E+T.$,)/+

goto(I_{11}, id)

I_9: $T \to id.(E)$,)/+
$T \to id.$,)/+

goto(I_{12}, E)

I_{13}: $T \to id(E.)$,)/+
$E \to E.+T$,)/+

goto(I_{12}, T)

I_8: $E \to T.$,)/+

goto(I_{12}, id)

I_9: $T \to id.(E)$,)/+
$T \to id.$,)/+

goto(I_{13},))

I_{14}: $T \to id(E).$,)/+

goto(I_{13}, +)

I_{11}: $E \to E+.T$,)/+
$T \to .id(E)$,)/+
$T \to .id$,)/+

CLR Parsing Table

States	ACTION +	id	(	)	$	GOTO T	E
I_0	S_4	S_3			~~1~~	2	1
I_1					Accept		
I_2	r_2				r_2		
I_3	r_4		S_5		r_4		
I_4		S_3				6	
I_5		S_9			~~7~~	8	7
I_6	r_1				r_1		
I_7	S_{11}			S_{10}			
I_8	r_2		~~r_2~~	r_2			
I_9	r_4		S_{12}	r_4			
I_{10}	r_3				r_3		
I_{11}		S_9				15 ~~8~~	
I_{12}		S_9				8	13
I_{13}	S_{11}			S_{14}			
I_{14}	r_3			r_3			
I_{15}	r_1			r_1			

Reduce

1. $E \rightarrow E + T$

~~$\rightarrow id$~~

2. $E \rightarrow T$
3. $T \rightarrow id(E)$
4. $T \rightarrow id$

I_1 : $E' \rightarrow E.\,, \$$

I_2 : $E \rightarrow T.\,, (\$, +)$

I_3 : $T \rightarrow id.\,, (\$, +)$

I_6 : $E \rightarrow E + T.\,, (\$, +)$

I_8 : $E \rightarrow T.\,, ( ),+)$

I_9 : $T \rightarrow id.\,, ( ),+)$

I_{10} : $T \rightarrow id(E).\,, (\$, +)$

~~I_8 : $T \rightarrow id($~~

I_{14} : $T \rightarrow id(E).\,, ( )/+)$

I_{15} : $E \rightarrow E + T.\,,)/+$